

## INFO


27/02/2002
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## AREAS

Mobile robotics · Marine robotics · Motion planning · Decision making · Obstacle detection and tracking · Control systems

## TOOLS

ROS · ROS 2 · Nav2 · Gazebo · ArduPilot · ArduSub · QGroundControl · LiDAR · Codesys · Modbus TCP · OPC-UA · MATLAB · Onshape · Docker · Linux · Git · LaTeX

## LANGUAGES

C++ · C · Python · Java

## SPOKEN

Portuguese (native)
English (fluent)
German (beginner)



# Francisco Pires da Silva

Robotics Engineer | Mobile & Marine Robotics

## Experience

### Research Assistant

**INESC TEC** – Centre for Robotics and Autonomous Systems (CRAS)
Motus Robotics group
Jan 2023 – Sep 2026 | Porto, Portugal

Worked on robotic vehicle navigation for offshore inspection across three EU-funded projects, on motion planning, obstacle detection and avoidance, control architecture and decision making. Accumulated 30+ days of field experience across trials and live demonstrations.

**AEROSUB** | Horizon Europe · automated offshore wind inspection
- Dynamic collision avoidance for close-range inspection, extending the velocity-obstacle work from the MSc dissertation to ASV interaction on offshore inspection routines.

**TALOS** | Horizon Europe · robotics and AI for photovoltaic inspection
- Developed the full software stack for the **LILY ASV**: motion planning, decision making, control architecture and UAV coordination.
- Demonstrated the fully autonomous mission at Barragem de Alqueva, Portugal.

**ATLANTIS** | H2020 · Atlantic offshore wind test platform
- Conceptualised and prototyped an active docking and retention system for UAV launch and recovery at sea.

## Education

### PhD Coursework, Electrical and Computer Engineering

Sep 2025 – Aug 2026 | University of Porto (FEUP)
Curricular year completed.

### Exchange Semester, Electrical and Computer Engineering

Oct 2024 – Mar 2025 | Technical University of Munich (TUM), Germany

### MSc in Electrical and Computer Engineering

Sep 2023 – Jul 2025 | University of Porto (FEUP) | GPA 18/20
Branch: Automation | Profile: Robotics and Autonomous Systems

**Dissertation:** *A Velocity Obstacle Avoidance Algorithm for Heterogeneous Robotic Vehicles* (19/20). A 3D velocity-obstacle algorithm that handles different vehicle kinematics under one method, letting holonomic and differential-drive surface vehicles and multirotor UAVs navigate through dynamic obstacles. Validated in simulation and in field trials on the NEST ASV at the ATLANTIS Test Centre.

### BSc in Electrical and Computer Engineering

Sep 2020 – Jul 2023 | University of Porto (FEUP) | GPA 18/20

## Courses, Training & Certifications

### MIT Portugal Marine Robotics Summer School

Jul 2026 – Jul 2026 | MIT Portugal, Azores, Portugal

### Marine Sensors for Ocean Environments, training course

Sep 2026 – Sep 2026 | INESC TEC / SINTEF, Porto, Portugal

### UAV Pilot Licence, EASA Open Category A1/A3

2024 | ANAC Portugal

**Francisco**
**Pires da Silva**

Robotics Engineer

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## Leadership & Outreach

### NEEEC-FEUP

Sep 2020 – Aug 2025 | Electrical and Computer Engineering Students' Branch, FEUP

- **Board Member, 2023/24**: head of the technological department, leading a team of 20 on projects and workshops.
- Logistics coordinator for NEEEIL-IT, a robotics competition.
- Organised and presented hands-on technical workshops for students on ROS 2, electronics, Arduino and Notion.
- Member of EESTEC, the European Association of Students of Electrical Engineering, since March 2022.

### Mentor, FEUP Mentorship Program

2022/23 | University of Porto (FEUP)

### Scouting, CNE Agrupamento 212 São Pedro

2008 – Aug 2020 | Vila Real, Portugal

## Selected Projects (more details on xizesse.github.io)

**Bird Recovery Detection** (2026 – present): YOLO26 and object-tracking vision pipeline tracking the recovery of rehabilitated birds, for the wildlife recovery centre at CRAS-HVUTAD.

**Hull Cleaning ROV** (2026, MIT Portugal MRSS): Adapted an ROV for detection and cleaning of invasive species carried on ship hulls, in the Azores.

**Christmas Market Methane Study** (2024, TUM): Mobile methane emission survey of the Munich Christmas markets, for the Environmental Sensing and Monitoring chair.

**Autonomous Quadruped Robot** (2023, FEUP): Walking quadruped on an RP2040, with IMU and time-of-flight obstacle detection.

**PID Ball Balancing Robot** (2023, NEEEC): Closed-loop control demonstrator built for the University of Porto fair, for high-school outreach.

**H.O.W.D.Y.** (2023): Networked multiplayer puzzle game in Java, where players cooperate from separate devices.

My portfolio website, with more about me and my projects:
**xizesse.github.io**